Core Business Objective

The primary goal of the BSWA is to transition a **tedious manual process** (currently managed via spreadsheets) into a **simple, semi-automated workflow**. The system is specifically designed to facilitate the creation of balanced, formatted alignment documents for high-complexity events like the **Olympic and Paralympic Games**.

Key Business Requirements

1. Data Integration & Automation

- **Automated Transcription:** Convert third-party input schedule data (spreadsheets) into a usable digital format.
- **Semi-Automated Workflow:** Replace manual data entry with a system that automates scheduling tasks while allowing for technician oversight.
- **Legacy Alignment:** The tool must integrate with the current CBC scheduling paradigm used for Para/Olympic alignment documents.

2. Schedule Optimization & Validation

- **Conflict Resolution:** The system must automatically detect and resolve scheduling overlaps or time conflicts where possible.
- **Automatic Load Balancing:** The application must distribute schedule material across **4 clusters of 8 nodes**.
- **Data Integrity:** Ensure that all modifications to the schedule adhere to fundamental broadcasting constraints through automated validation.

3. User Experience & Efficiency

- **Visual Interface:** Provide a graphical, interactive display (the "scheduling matrix") to make data manipulation easier than traditional spreadsheets.
- **Drag-and-Drop Interaction:** Support intuitive user inputs like mouse clicks and drag-and-drop for modifying schedules.
- **Real-time Feedback:** Updates made by technicians must be reflected visually and in memory immediately.

4. Operational Constraints & Scope

- **No Infrastructure Changes:** The solution must work within CBC's existing infrastructure; the project does not provide new hardware or post-deployment maintenance.
- **Containerized Deployment:** The final product must be delivered as a **Docker image** for flexible deployment by the CBC.
- **Standardized Input:** The business relies on third-party data arriving in a consistent, pre-formatted tabular structure, as the tool does not include raw data cleaning features.